Journal of Cardiology 81 (2023) 469–475

Contents lists available at ScienceDirect

Journal of Cardiology

journal homepage: www.elsevier.com/locate/jjcc

Original Article

Prognostic impact of plasma xanthine oxidoreductase activity in patients with heart failure with atrial fibrillation

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ARTICLE INFO

Article history:
Received 20 November 2022
Received in revised form 2 February 2023
Accepted 7 February 2023
Available online 21 February 2023

Keywords:
Xanthine oxidoreductase
Heart failure
Atrial fibrillation

ABSTRACT

Background: Xanthine oxidoreductase (XOR) is a rate-limiting enzyme for uric acid (UA) production and plays an important role in generating reactive oxygen species (ROS). Overproduction of ROS is reported to contribute to the pathophysiology of atrial fibrillation (AF), however, the prognostic impact of plasma XOR activity in patients with heart failure (HF) with AF is undetermined.
Methods: We measured plasma XOR activity in 475 HF patients, including those with sinus rhythm (HF-SR, $n = 211$), and those with AF (HF-AF, $n = 264$). The type of AF included paroxysmal ($n = 128$) and persistent ($n = 136$) AF. All patients were prospectively followed up for a median period of 804 days.
Results: HF-AF patients had significantly higher plasma XOR activity and serum UA levels compared with HF-SR patients. Both plasma XOR activity and serum UA levels were higher in patients with persistent AF than in those with SR and with paroxysmal AF. Multivariate linear regression analysis showed that persistent AF was independently associated with increased XOR activity. During the follow-up period, there were 79 major adverse cardiovascular events (MACEs). HF-AF patients with MACEs had higher plasma XOR activity compared with those without MACEs, while there were no significant differences in serum UA levels. Multivariate Cox proportional analysis showed that high XOR activity was an independent risk factor for MACEs after adjustment for confounding factors. Kaplan-Meier analysis revealed that the high XOR activity group had a higher risk of MACEs than the low XOR activity group. The prediction model was significantly improved by the addition of XOR activity to the basic predictors.
Conclusions: HF-AF patients had significantly higher plasma XOR activity compared with HF-SR patients. Plasma XOR activity proved to be a reliable indicator for MACEs in HF-AF patients.

Introduction

An increasing prevalence of atrial fibrillation (AF) and heart failure (HF) is a worldwide public health issue in the aging population [1]. AF and HF frequently co-exist [2,3], and the occurrence of one contributes to the development of the other [4]. AF itself leads to cumulative electrophysiological and structural atrial remodeling [5]. A loss of atrial kick has been shown to reduce blood flow volume by approximately 10 %, contributing to the development of HF [6]. Among the various mechanisms underlying AF association with HF, reactive oxygen species (ROS) is considered to be one of the major contributory mechanisms [7]. AF reportedly induces ROS production, which in part results in increased xanthine oxidoreductase (XOR) activity [8]. XOR is a rate-limiting enzyme in purine metabolism, and catalyzes the oxidation of hypoxanthine to xanthine, and xanthine to uric acid (UA), and also generates the ROS [9,10]. We previously reported that plasma XOR activity is associated with clinical outcomes in HF [10,11]. However, the prognostic impact of plasma XOR activity in patients with HF with AF (HF-AF) has not yet been elucidated. The aim of this study was to assess the relationship of plasma XOR activity with AF, and its prognostic power in patients with HF-AF.

Methods

Study subjects

The present study enrolled 475 consecutive patients who were admitted to our hospital for the diagnosis and/or treatment of HF. The clinical diagnosis of HF was based on the Framingham criteria, which requires the presence of either two major criteria, or one major and two

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https://doi.org/10.1016/j.jjcc.2023.02.003

minor criteria [12]. The patients were divided into two groups: those with and without AF, i.e. HF with sinus rhythm (HF-SR) and HF-AF. The type of AF included paroxysmal and persistent AF. Paroxysmal AF is defined as an episode of AF that terminates spontaneously, or with intervention, within seven days, and persistent AF is defined as an episode of AF that is sustained for more than seven days [13]. Persistent AF includes cases of longstanding AF (more than one year). Both patients with a first diagnosis of AF at admission or during hospitalization using resting 12-lead electrocardiogram and/or electrocardiogram monitoring and those with a history of AF were included. The present study excluded patients who presented with acute coronary syndrome (ACS) within the three months prior to admission, and those with active hepatic disease, pulmonary disease, and/or malignant disease. The flowchart of enrollment of heart failure patients is shown in Fig. 1. Clinical data including age, gender, New York Heart Association (NYHA) functional class, and medical history, such as hypertension, diabetes mellitus, and hyperlipidemia, were collected from the patients' medical records or history of medical therapy. Medications were assessed at discharge. The investigation conforms with the principles outlined in the *Declaration of Helsinki*. The study protocol was approved by the institutional ethics committee of Yamagata University School of Medicine, and all patients provided their written informed consent for study participation.

Xanthine oxidoreductase activity assay

Venous blood samples were collected in the early morning within 24 h after admission, and centrifuged at 3000*g* for 15 min at 4 °C, whereupon the plasma was stored at −80 °C until analysis. The XOR activity assay was performed using stable, isotope-labeled substrate and liquid chromatography triple quadrupole mass spectrometry (Sanwa Kagaku Kenkyusho Co., Ltd., Nagoya, Japan) [14].

Endpoint and follow-up period

All patients were prospectively followed up for a median period of 804 days (IQR 441–1447 days). The endpoints were major adverse cardiovascular events (MACEs), including re-hospitalization for HF, ACS, and cardiac death (defined as death due to progressive HF, ACS, or sudden cardiac death). Re-hospitalization for HF was defined as worsening symptoms of HF that requires urgent admission.

Statistical analysis

The results are expressed as the mean ± standard deviation (SD) for continuous variables, and percentages for categorical variables. Skewed values are presented as the median and IQR. We used *t*-tests and chi-square tests to compare continuous and categorical variables, respectively. If the data were not normally distributed, the Mann-Whitney *U* test was employed. Differences among groups were analyzed by analysis of variance. Comparison of plasma XOR activity among the three groups was performed using the Kruskal-Wallis test. The factors associated with the increase in plasma XOR activity were analyzed by linear regression analysis. Cox's proportional hazard analysis was used to determine the independent predictors for MACEs. Significant variables from the univariate analysis were then entered into the multivariate analysis. Event-free survival curves were constructed according to the Kaplan-Meier method and compared using log-rank tests. We calculated the net reclassification index (NRI) and integrated discrimination index (IDI) to estimate the level of improvement in the predictive ability of the baseline model following the addition of plasma XOR activity. Values of $p < 0.05$ were considered statistically significant. All statistical analyses were performed with a standard software package (JMP version 12; SAS Institute, Cary, NC, USA).

Results

Baseline clinical characteristics of HF-SR and HF-AF patients

There were 211 HF-SR patients and 264 HF-AF patients. The HF-AF group comprised 128 paroxysmal and 136 persistent AF patients. A comparison of the baseline clinical characteristics of the HF-SR and HF-AF patients is shown in Table 1. The HF-AF patients had a

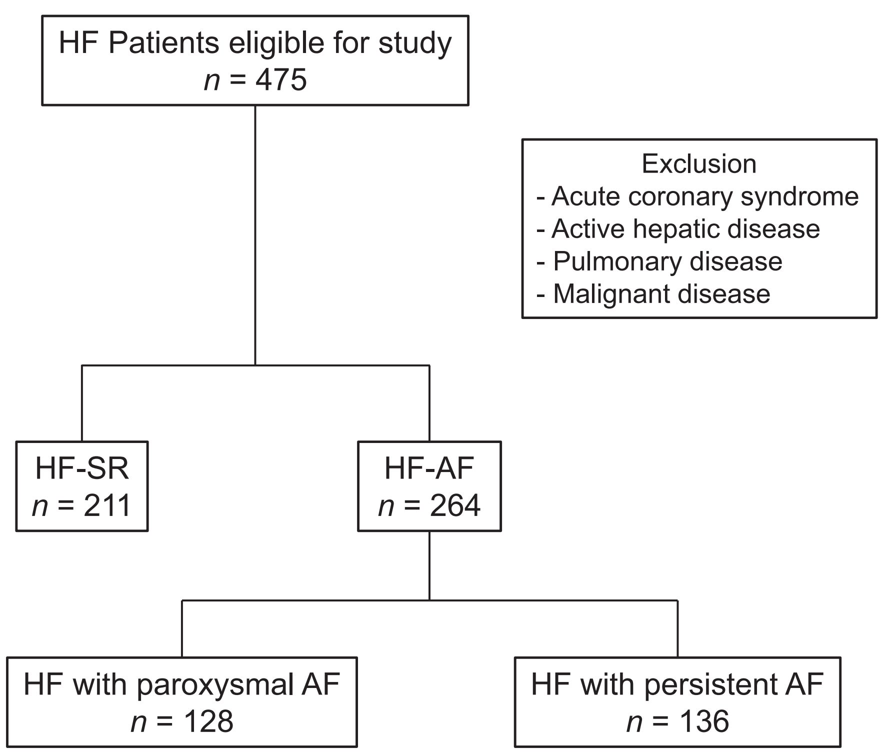

Fig. 1. The flowchart of enrollment of heart failure patients.
AF, atrial fibrillation; HF, heart failure; SR, sinus rhythm.

Table 1
Comparison of the baseline clinical characteristics of HF-SR and HF-AF patients.

Variables	HF-SR $n = 211$	HF-AF $n = 264$	p-Value
Age (years)	69 ± 12	70 ± 12	0.357
Male, n (%)	120 (57)	165 (63)	0.214
BMI (kg/m^2)	22.6 ± 4.1	21.9 ± 3.8	0.077
Hypertension, n (%)	167 (79)	208 (79)	0.924
Dyslipidemia, n (%)	58 (27)	68 (26)	0.671
Diabetes mellitus, n (%)	63 (30)	63 (24)	0.142
NYHA III-IV, n (%)	83 (39)	120 (45)	0.180
Etiology			
IHD/non-IHD/VHD/others	54/91/35/31	50/114/57/43	0.253
Echocardiographic data			
LVEDD (mm)	55 ± 10	55 ± 10	0.801
LVEF (%)	48 ± 18	50 ± 17	0.337
LAD (mm)	41 ± 6	47 ± 9	<0.001
Blood examination			
eGFR (mL/min/1.73 m^2)	64 ± 26	66 ± 24	0.518
BNP (pg/mL)	305 (95–852)	347 (159–640)	0.404
UA (mg/dL)	6.0 ± 1.0	6.6 ± 2.4	0.003
XOR (pmol/h/mL)	64 (42–102)	75 (42–144)	0.026
Hemodynamic data			
SBP (mmHg)	120 ± 22	115 ± 28	0.436
DBP (mmHg)	69 ± 16	63 ± 16	0.130
Heart rate (bpm)	69 ± 18	74 ± 21	0.265
Medications at discharge			
ACEIs and/or ARBs, n (%)	152 (72)	191 (72)	0.940
β-Blockers, n (%)	131 (62)	185 (70)	0.067
XOR inhibitors, n (%)	17 (8)	28 (11)	0.343
Statins, n (%)	55 (26)	55 (21)	0.180
Antiarrhythmic drugs, n (%)			
Class I	N/A	19 (7)	N/A
Class III	N/A	21 (8)	N/A
Digoxin	N/A	46 (17)	N/A

Data are expressed as mean ± SD, number (percentage), or median (interquartile range). ACEIs, angiotensin-converting enzyme inhibitors; AF, atrial fibrillation; ARBs, angiotensin II receptor blockers; BMI, body mass index; BNP, B-type natriuretic peptide; HF, heart failure; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; IHD, ischemic heart disease; LAD, left atrial diameter; LVEDD, left ventricular end-diastolic diameter; LVEF, left ventricular ejection fraction; NYHA functional class, New York Heart Association functional class; SBP, systolic blood pressure; SR, sinus rhythm; UA, uric acid; VHD, valvular heart disease; XOR, xanthine oxidoreductase.

significantly larger left atrial diameter (LAD) than the HF-SR patients, while left ventricular ejection fraction and end-diastolic diameter were similar between the groups. There were no significant differences in age, gender, NYHA functional class, medical history, and etiology of HF between the groups. The HF-AF patients had significantly higher plasma XOR activity and serum UA levels compared with the HF-SR patients. Furthermore, both plasma XOR activity and serum UA levels were higher in the patients with persistent AF than in those with SR and with paroxysmal AF (Fig. 2).

The factors associated with the increase in plasma XOR activity

The univariate linear regression analysis demonstrated that gender ($p = 0.045$), persistent AF ($p = 0.002$), and UA levels ($p = 0.044$) were associated with increased XOR activity. Multivariate analysis showed that only persistent AF ($p = 0.001$) was independently associated with increased XOR activity. Paroxysmal AF ($p = 0.057$) and B-type natriuretic peptide (BNP) level ($p = 0.062$) tended to be associated with increased XOR activity, but not significantly (Table 2).

Prognostic impact of plasma XOR activity in HF, with and without AF

During the median follow-up period of 804 days, there were 79 MACEs in the HF-AF patients. The MACEs included re-hospitalization for HF in 37 patients, ACS in 8 patients, and cardiac death in 34 patients. A comparison of the clinical characteristics of HF-AF patients, with and without MACEs, is shown in Table 3. The HF-AF patients with MACEs had a more severe NYHA functional class, larger LAD, lower levels of estimated glomerular filtration rate (eGFR), higher levels of BNP, and lower use of statins than those without MACEs. The HF-AF patients with MACEs had higher plasma XOR activity compared with those without MACEs, while there were no significant differences in serum UA levels between those with and without MACEs. To examine the risk factors for MACEs, we performed univariate and multivariate Cox proportional hazard regression analyses. As shown in Table 4, the univariate Cox proportional hazard regression analysis revealed that patient age, NYHA functional class, eGFR, BNP, and plasma XOR activity, were significantly associated with MACEs. The multivariate analysis showed that the plasma XOR activity and NYHA functional class were independent risk factors for MACEs after adjustment for confounding factors.

All the patients were divided into two groups according to median value of the plasma XOR activity (75 pmol/h/mL). Kaplan-Meier analysis revealed that the high XOR activity group had a higher risk of MACEs than the low XOR activity group (Fig. 3). We investigated whether the model fit and discrimination were improved by the addition of plasma XOR activity to the basic predictors of age, NYHA functional class,

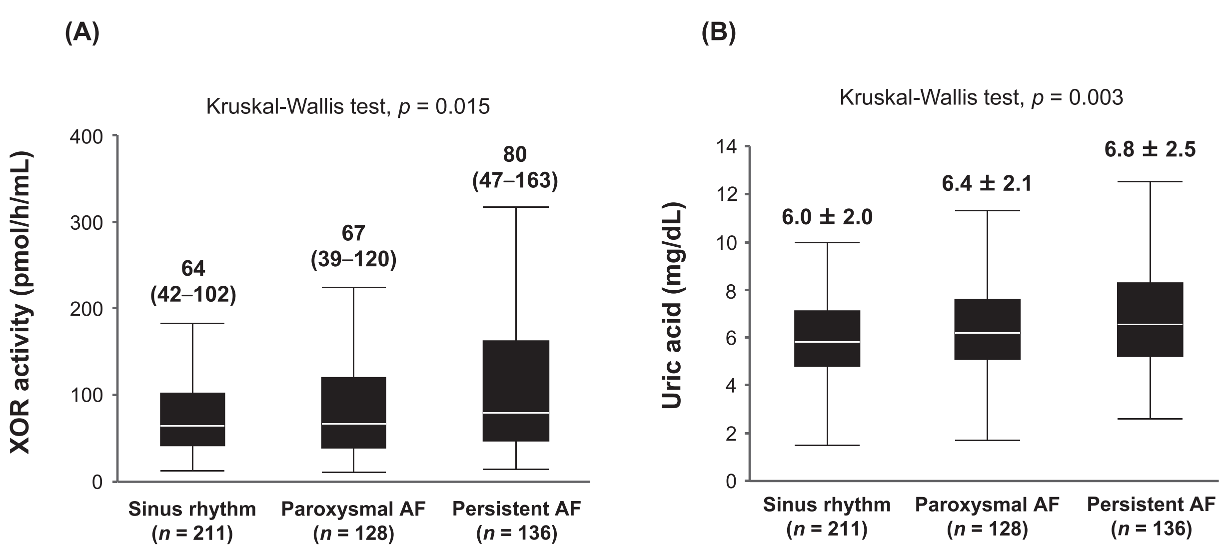

Fig. 2. (A) The xanthine oxidoreductase (XOR) activity in the patients with paroxysmal atrial fibrillation (AF) was greater than in those with sinus rhythm. This activity was further increased in those patients with persistent AF ($p = 0.015$). (B) Uric acid levels in patients with paroxysmal AF were greater than in those with sinus rhythm, and were further increased in those patients with persistent AF ($p = 0.003$).

Table 2
The factors associated with the increase in plasma XOR activity.

	Univariate analysis				Multivariate analysis			
	Standardized coefficient (beta)	R^2	t	*p*-Value	Standardized coefficient (beta)	R^2	t	*p*-Value
Age	−0.054	0.003	−1.19	0.233				
Male gender	0.092	0.009	2.01	0.045	0.026	<0.001	0.48	0.630
BMI	0.043	0.002	0.79	0.432				
Hypertension	0.046	0.002	1.01	0.315				
Dyslipidemia	0.079	0.006	1.71	0.087				
Diabetes mellitus	0.068	0.005	1.50	0.134				
NYHA III-IV	0.075	0.006	1.63	0.103				
Paroxysmal AF vs. SR	0.103	0.011	1.91	0.057				
Persistent AF vs. SR	0.164	0.027	3.09	0.002	0.180	0.032	3.33	0.001
eGFR	<0.001	<0.001	0.01	0.916				
BNP	0.084	0.007	1.82	0.062				
UA	0.115	0.013	2.51	0.044	0.039	0.001	0.69	0.489

AF, atrial fibrillation; BMI, body mass index; BNP, B-type natriuretic peptide; eGFR, estimated glomerular filtration rate; NYHA functional class, New York Heart Association functional class; SR, sinus rhythm; UA, uric acid; XOR, xanthine oxidoreductase.

eGFR, and log BNP. As shown in Fig. 4, the C index was significantly improved by the addition of plasma XOR activity to the baseline model. In addition, both the NRI and IDI were significantly improved by the addition of plasma XOR activity to the basic predictors (NRI 0.524; 95 % confidence interval 0.278–0.770; $p < 0.001$ and IDI 0.090; confidence interval 0.045–0.136; $p < 0.001$).

Table 3
Comparison of the clinical characteristics of HF-AF patients, with and without MACEs.

Variables	HF-AF ($n = 264$)		*p*-Value
	Event-free	MACEs	
	$n = 185$	$n = 79$	
Age (years)	70 ± 12	72 ± 12	0.161
Male, n (%)	111 (60)	54 (68)	0.199
BMI (kg/m^2)	22.0 ± 4.1	21.6 ± 3.1	0.548
Hypertension, n (%)	143 (77)	65 (82)	0.359
Dyslipidemia, n (%)	50 (27)	13 (16)	0.058
Diabetes mellitus, n (%)	45 (24)	23 (29)	0.419
NYHA III-IV, n (%)	73 (39)	47 (59)	0.003
Etiology			
IHD/non-IHD/VHD/Others	31/79/42/33	19/35/15/10	0.417
Echocardiographic data			
LVEDD (mm)	55 ± 10	56 ± 10	0.433
LVEF (%)	50 ± 17	49 ± 17	0.616
LAD (mm)	46 ± 9	50 ± 9	0.006
Blood examination			
eGFR (mL/min/1.73 m^2)	68 ± 25	61 ± 23	0.030
BNP (pg/mL)	277 (151–581)	434 (288–928)	0.020
UA (mg/dL)	6.5 ± 2.1	6.9 ± 2.8	0.185
XOR (pmol/h/mL)	69 (39–111)	116 (50–276)	<0.001
Hemodynamic data			
SBP (mmHg)	114 ± 27	118 ± 30	0.237
DBP (mmHg)	63 ± 16	64 ± 18	0.616
Heart rate (bpm)	75 ± 20	75 ± 24	0.922
Medications at discharge			
ACEIs and/or ARBs, n (%)	133 (72)	58 (73)	0.799
β-Blockers, n (%)	132 (71)	53 (67)	0.491
XOR inhibitors, n (%)	20 (11)	8 (10)	0.868
Statins, n (%)	46 (25)	9 (11)	0.010
Antiarrhythmic drugs, n (%)			
Class I	14 (8)	5 (6)	0.751
Class III	15 (8)	6 (8)	0.909
Digoxin	31 (17)	15 (19)	0.711

Data are expressed as mean ± SD, number (percentage), or median (interquartile range). ACEIs, angiotensin-converting enzyme inhibitors; AF, atrial fibrillation; ARBs, angiotensin II receptor blockers; BMI, body mass index; BNP, B-type natriuretic peptide; HF, heart failure; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; IHD, ischemic heart disease; LAD, left atrial diameter; LVEDD, left ventricular end-diastolic diameter; LVEF, left ventricular ejection fraction; MACEs, major adverse cardiovascular events; NYHA functional class, New York Heart Association functional class; SBP, systolic blood pressure; UA, uric acid; VHD, valvular heart disease; XOR, xanthine oxidoreductase.

Discussion

Main findings

The main findings of this study were as follows: (1) HF-AF patients had significantly higher plasma XOR activity compared with HF-SR patients; (2) persistent AF was independently associated with the increase in the plasma XOR activity; (3) high plasma XOR activity was a risk factor for MACEs in HF-AF patients; and (4) the prognostic capacity of the prediction model for HF-AF patients was improved with the inclusion of plasma XOR activity.

The vicious cycle between ROS and AF

Excessive production of ROS is often associated with the pathophysiology of AF. Elevated ROS directly contribute to the structural and electrical remodeling of atria, and lead to the development of AF [15]. ROS-induced endothelial dysfunction is a further underlying mechanism of AF [16]. Endothelial dysfunction results in decreased NO bioavailability and impaired vasodilation, leading to a left atrial afterload that triggers AF [17].

XOR consists of two interconvertible forms: xanthine dehydrogenase (XDH), and xanthine oxidase (XO) [18]. Although XDH catalyzes the hydroxylation of xanthine to UA with the consumption of nicotinamide adenine dinucleotide (NAD), XO preferentially consumes molecular oxygen and produces the ROS, such as superoxide anion (O_2^-) [19].

Table 4
Univariate and multivariate Cox proportional hazard analyses for predicting MACEs in HF-AF patients.

Variables	Univariate analysis			Multivariate analysis		
	HR	95 % CI	*p*-Value	HR	95 % CI	*p*-Value
Age[a]	1.278	1.011–1.642	0.040	1.137	0.888–1.457	0.300
Male gender	1.295	0.815–2.116	0.279			
BMI[a]	0.910	0.685–1.194	0.503			
Hypertension	1.242	0.718–2.306	0.453			
Dyslipidemia	0.613	0.323–1.073	0.089			
Diabetes mellitus	1.314	0.793–2.106	0.281			
NYHA (III/IV vs. II)	2.362	1.511–3.740	<0.001	1.797	1.117–2.920	0.016
LAD[a]	1.285	0.967–1.696	0.084			
eGFR[a]	0.690	0.528–0.890	0.004	0.841	0.638–1.108	0.205
Log BNP[a]	1.466	1.164–1.856	0.001	1.207	0.941–1.547	0.135
UA[a]	1.213	0.957–1.525	0.109			
XOR[a]	1.433	1.244–1.611	<0.001	1.395	1.215–1.600	<0.001
Statins use	0.548	0.254–1.043	0.069			

AF, atrial fibrillation; BMI, body mass index; BNP, B-type natriuretic peptide; HF, heart failure; CI, confidence interval; eGFR, estimated glomerular filtration rate; LAD, left atrial diameter; MACEs, major adverse cardiovascular events; NYHA functional class, New York Heart Association functional class; UA, uric acid; XOR, xanthine oxidoreductase.
[a] Per 1-SD increase.

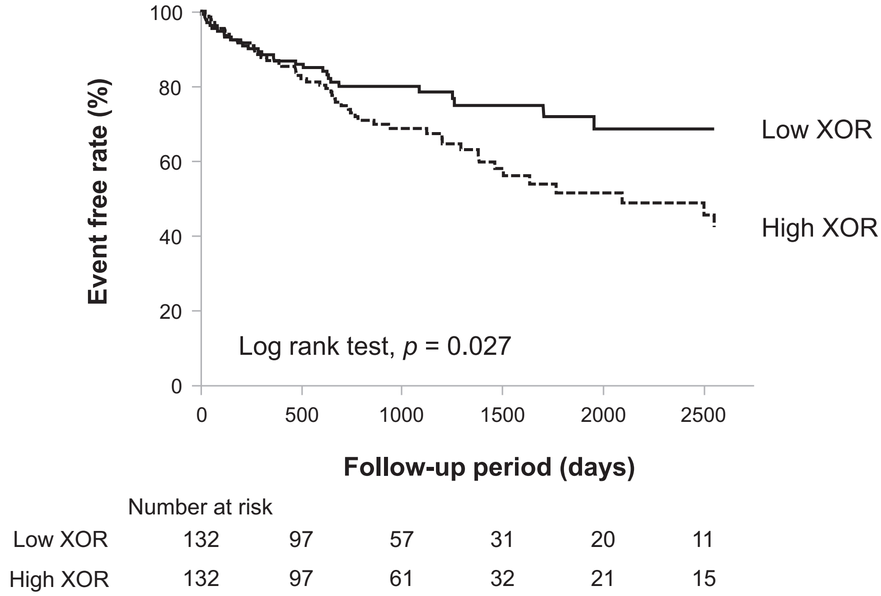

Fig. 3. Kaplan-Meier analysis of the time to subsequent outcome in patients with heart failure and atrial fibrillation. The major adverse cardiovascular events rate was higher in patients with high xanthine oxidoreductase (XOR) activity than in those with low XOR activity.

The XO-derived O_2^- radical can react with NO to form peroxynitrate ($ONOO^-$), which is a powerful oxidizing mediator of vascular endothelial injury [20]. Furthermore, $ONOO^-$ can mediate the conversion of XDH to XO, leading to further ROS production [21]. XOR has been implicated as an important source of ROS in the vascular system [22]. Recently we reported a significant association between plasma XOR activity and coronary artery spasm [23], and HF with preserved ejection fraction [11], which are considered to be closely associated with endothelial dysfunction.

Conversely, in a swine model where AF was induced by rapid atrial pacing, the AF increased superoxide production in both the left atrium and left atrial appendage, and increased XO activity contributed to the superoxide elevation in left atrial appendage [8]. The authors concluded that this increase in superoxide and its reactive metabolites may contribute to

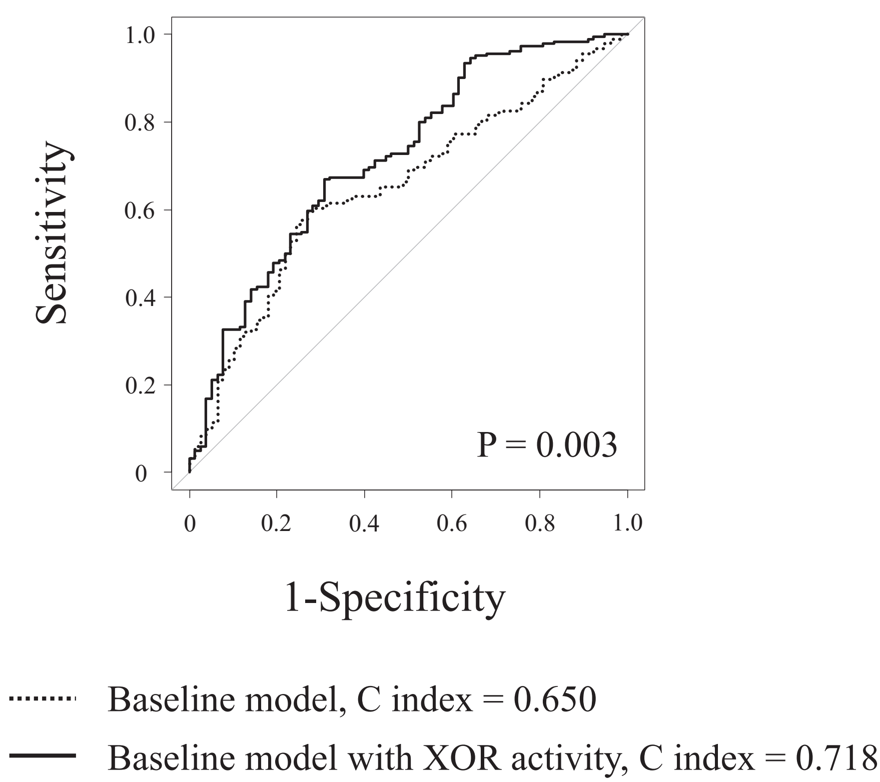

Fig. 4. Comparison of receiver operating curve analysis of major adverse cardiovascular events. The model fit and discrimination were improved by the addition of plasma xanthine oxidoreductase (XOR) activity to the basic predictors of age, New York Heart Association functional class, estimated glomerular filtration rate, and log B-type natriuretic peptide. The C index was significantly improved by the addition of plasma XOR activity to the baseline model.

the pathophysiological consequences of AF, such as inflammation and tissue remodeling [8]. Similarly, in the fibrillating human atrial myocardium, the increased superoxide production was observed [24]. Thus, AF itself can induce systemic oxidative stress, inflammation, and subsequent endothelial dysfunction [25,26]. Indeed, in our study plasma XOR activity was elevated in patients with paroxysmal AF, and was even greater in those with persistent AF. Furthermore, linear regression analysis showed that paroxysmal AF tended to be associated with increased XOR activity and persistent AF was independently associated with XOR elevation. These results suggest that XOR-derived ROS and AF aggravate each other, and appear to form a vicious cycle.

XOR activity in patients with HF-AF

Both AF and HF facilitate the occurrence, and aggravate the prognosis, of each other. These diseases frequently co-exist, have many common risk factors, and are associated with increased risk of cardiovascular events [27]. Several biomarkers, reflecting cardiac volume overload, myocardial damage, inflammation, and renal impairment have been shown to be associated with risk of cardiovascular events in patients with HF-AF [28]. Oxidative stress is the excessive accumulation of ROS relative to antioxidant activity and is a major cause of cardiovascular disease in animal models [29]. Previous observations support the hypothesis that ROS plays a key role in human HF, although to date the clinical data have been less than compelling, and the role of ROS has not been definitively established. This is in part due to the difficulty in determining ROS activity in vivo, and that HF patients often have confounding comorbidities that could induce alterations in biochemical markers [30].

XOR is recognized to play an important role in the vascular oxidative stress underlying endothelial dysfunction [19,22]. Inhibition of XOR activity improves endothelial function by reducing vascular oxidative stress in HF patients [31,32]. These results suggest that XOR, via its effect on endothelial dysfunction, could be a key modulator of HF progression. In the present study, high plasma XOR activity was associated with adverse cardiovascular events in HF-AF patients. In addition, there was a significant relationship between increased plasma XOR activity and MACEs in HF-AF patients. Despite the fact that no differences, other than LAD, were found in the baseline characteristics between the HF-AF and HF-SR patients, XOR activity was significantly higher in the HF-AF patients. As the results of linear regression analysis, persistent AF was associated with elevated XOR. BNP also tended to be associated with elevated XOR level. These findings imply that not only the presence of AF itself, but also coexisting HF may be an important cause of progressive increase of XOR activity. The vicious cycle between ROS and AF may promote a greater increase of XOR activity in HF-AF patients. Elevated plasma XOR activity is not just a biomarker, but may contribute to the worsening of the condition in patients with HF-AF.

Limitations

The current study had several limitations. First, plasma XOR activity was measured once on the day of admission and XOR inhibitors were administered after measurement of XOR activity. The effect of medications administered during hospitalization on XOR activity was not investigated. Second, further studies are needed to determine whether XOR inhibitors improve the prognosis of HF-AF patients. Third, the endothelial function was not measured, thus, the association between endothelial dysfunction and plasma XOR activity was not evaluated in patients with HF-AF. Finally, the effect of AF catheter ablation on XOR activity could not be investigated in the present study.

Conclusions

The present study revealed that HF-AF patients had significantly higher plasma XOR activity compared with HF-SR patients. High plasma XOR activity was associated with MACEs in HF-AF patients. Plasma XOR activity proved to be a reliable indicator for MACEs in HF-AF patients.

Declaration of competing interest

None declared.

Acknowledgments

This work was in part supported by the consigned research fund from Sanwa Kagaku Kenkyusho Co. Ltd.

Funding

None.

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